

●Low flow controls on benthic and hyporheic macroinvertebrate assemblages during supra-seasonal drought

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Abstract:

Despite the widely accepted importance of the hyporheic zone as a habitat for stream macroinvertebrates during floods, few data exist regarding community composition and distribution during periods of low flow or drought in perennial streams. Integrating research on hyporheic invertebrates with results from a long-term study of a UK river provided the opportunity to examine how surface and hyporheic macroinvertebrate communities respond to inter-annual variability in river flow and periods of groundwater drought. Changes in the riverine macroinvertebrate community associated with low flow included a reduction in species richness and the number of individuals per sample, particularly aquatic insects. The hyporheic community was characterized by a relatively homogeneous composition during a period of severe low flow, punctuated by short-term changes associated with variation in water temperature rather than changes in discharge. We present a conceptual model of the processes influencing benthic and hyporheic invertebrates under low-flow conditions. Previous studies have seldom integrated these two assemblages and their interactions. The model presented highlights the potential importance of surface water and hyporheic zone linkages for riverine invertebrate communities under a range of flow conditions. Copyright © 2009 John Wiley & Sons, Ltd.

KEY WORDS benthos; hyporheos; low flows; drought; hyporheic processes; invertebrates; groundwater

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INTRODUCTION

Natural low flows associated with droughts originate from a deficit of precipitation (Smakhtin, 2001). Droughts can occur in almost any biogeographical setting, although the onset of an individual event can be difficult to determine (Humphries and Baldwin, 2003; Smakhtin and Schipper, 2008). Following an initial deficit of precipitation (meteorological drought), river discharge and water levels in other surface water bodies decline leading to 'hydrological drought' within the drainage basin and/or wider region. Ultimately, without sufficient meteorological input (recharge), groundwater levels within aquifers will decline, resulting in a 'groundwater drought', the impact of which may be compounded by anthropogenic water resource requirements for agricultural, industrial and domestic uses (Tallaksen and van Lanen, 2004).

River flow regime variability and low flows associated with drought conditions have been widely studied in lotic systems (Smakhtin, 2001), and their role in structuring in-stream communities is now recognized (Lytle and Poff, 2004; Monk *et al.*, 2008). However, due to the complexities of defining and determining the onset of events, hydroecological data documenting responses of in-stream communities to droughts, from their onset to

recovery, are relatively limited compared with studies reporting the ecological responses to floods (Suren and Jowett, 2006; Lake, 2007). In addition, there are marked differences in the manifestation of drought between individual catchments and our understanding of how in-stream ecological communities respond varies regionally (Demuth and Young, 2004).

The response of in-stream organisms to drought largely reflects the predictability and severity, including the duration, of the event (Lake, 2003). The greatest understanding of the role of drought within lotic ecosystems is for those subject to predictable 'seasonal' droughts in semi-arid environments (Acuna *et al.*, 2005; Bonada *et al.*, 2006). Those communities experiencing regular drought, typical of Mediterranean environments, frequently display behavioural and physiological adaptations that enable them to withstand prolonged low flows or cessation of flow (Bonada *et al.*, 2006). Ecological data available for droughts within temperate environments are comparatively limited (Wood and Armitage, 2004; Lake, 2007). Aquatic invertebrate communities in temperate zone perennial lotic ecosystems subject to irregular and/or high magnitude events are seldom adapted to withstand the extreme conditions and, as a result, are usually severely impacted when flow declines or ceases (Wright and Berrie, 1987; Caruso, 2002; Lake, 2007).

Droughts are 'ramp disturbances' (*sensu* Lake, 2003) that gradually increase in intensity over time. The

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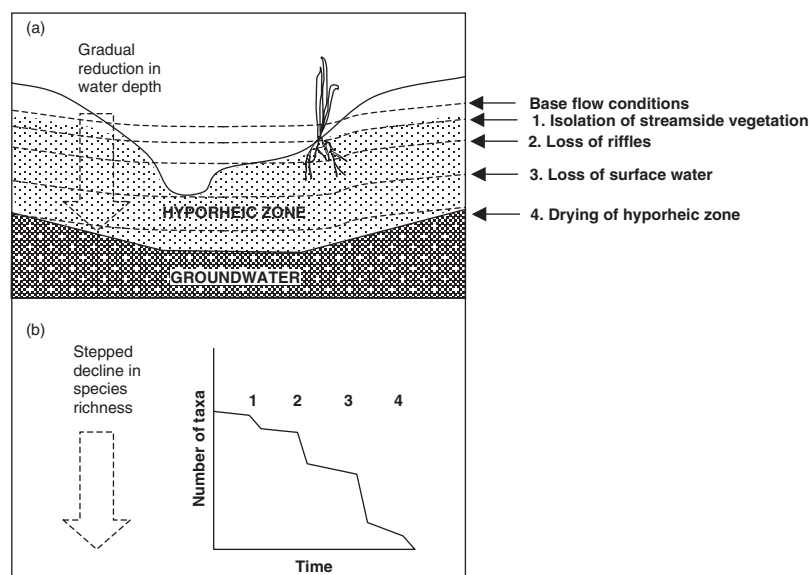


Figure 1. Changes in river stage and macroinvertebrate assemblage composition associated with supra-seasonal drought: (a) cross-section of a conceptualized channel during critical stages of drying; (b) hypothesized 'stepped' changes in species richness corresponding to these critical stages (adapted from Boulton, 2003)

response of lotic communities to drought and reduced river discharge has been characterized by gradual (ramp) changes punctuated by significant 'stepped' responses as thresholds between critical levels are crossed (Boulton, 2003). These steps reflect the gradual reduction in river stage (water depth) coinciding with ecologically significant threshold changes in discharge or the exposure of particular habitats. Examples include the isolation of streamside vegetation, cessation of flow, the isolation of surface water into pools, loss of surface water and far less studied the decline or loss of free water within the hyporheic zone (Figure 1).

The functional significance of the hyporheic zone has gained increasing prominence in recent years (Boulton *et al.*, 1998; Boulton, 2007). This reflects the recognition that the habitat supports a number of unique (obligate) taxa and has wider linkages in the landscape with other surface and groundwater habitats (Malard *et al.*, 2002). There is increasing evidence that processes operating within the hyporheic zone may significantly contribute to maintaining ecosystem health (Tomlinson *et al.*, 2007; Pinay *et al.*, 2009) through the provision of key ecosystem services (Boulton *et al.*, 2008). It is now widely recognized that the hyporheic zone is a focal point for important biogeochemical processes and the transient storage of nutrients (Mulholland *et al.*, 2008; Pinay *et al.*, 2009). In addition, the exchange of water within the hyporheic zone may locally influence dissolved oxygen concentrations, thermal properties and sedimentary characteristics required to support salmonid fisheries (Malcolm *et al.*, 2005).

In this article, we synthesize the existing hydroecological data available for the Little Stour River (Kent, UK) to examine macroinvertebrate community responses to river flow variability and drought-related low flows. The benthic macroinvertebrate hydroecology of the river has been

extensively studied for over a decade in relation to flow variability, in particular the influence of low flows associated with droughts (Wood and Petts, 1999; Wood *et al.*, 2000; Wood and Armitage, 2004). We present data from a long-term study of inter-annual variability of the benthic community (1992–1999) and data collected as part of a detailed monthly investigation of the benthic and hyporheic invertebrate communities during a groundwater drought in 2006. In particular, we assess whether the benthic and hyporheic fauna respond similarly to drought and whether there is evidence of marked 'step responses' to the ramp disturbance of drought in the hyporheic zone where effects of drying are hypothesized to be buffered by the saturated sediments. These results are used along with other published information to develop a conceptual model to demonstrate how interactions between surface and groundwater influence hydrological processes within the hyporheic zone which, in turn, may structure habitat availability and the benthic and hyporheic zone communities.

STUDY SITE

The Little Stour River (Kent, UK) is a small lowland chalk stream, 11.5-km long, draining a catchment area of approximately 213 km² (51.275°N 1.168°E). The highly permeable nature of the catchment results in a low drainage density, which is typical of groundwater-dominated streams. The sedimentary calcareous rocks result in relatively high conductivities (*ca* 580 $\mu\text{S cm}^{-1}$). Mean annual precipitation within the catchment is *ca* 650 mm per year (Wood and Petts, 1994). The river is usually perennial below the spring head, although a 1-km reach has been dewatered on three previous occasions in the last century during supra-seasonal drought events (1949, 1991–1992 and 1996–1997), with the latter two

events being studied in detail (Wood and Armitage, 2004). A subsequent drought event impacted the site and much of southern England between 2004 and 2006 (Marsh, 2007), although the Little Stour maintained perennial flow along its entire length throughout this period.

METHODS

The macroinvertebrate community of the river was sampled annually (1992–1999) from nine sites along the upper river. Macroinvertebrates were sampled during base flow conditions (late August–early September) using a semi-quantitative kick-sampling technique over a 2-min period (Wood and Armitage, 2004). During 2006, both benthic and hyporheic invertebrate communities were sampled from four riffle sites on the river between April 2006 and October 2006. This coincided with the latter stages of a supra-seasonal drought resulting from below-average rainfall between November 2004 and June 2006 (Marsh *et al.*, 2007). For further details of site locations and physical characteristics, including relative flow permanence, see Wood and Petts (1999) and Wood *et al.* (2000).

During the intensive study in 2006, five benthic samples were collected at each of the four riffle sites on the upper river each month using a Surber sampler (0.1 m², 250-µm mesh net) over a 30-s period, disturbing the substratum to a depth of 50 mm. Associated with each benthic sample, hyporheic invertebrate samples were collected from 20-cm deep polyvinylchloride (PVC) wells (25-mm internal diameter) following the procedure outlined by Boulton and Stanley (1995). PVC wells were inserted into the riverbed using a stainless steel bar and samples can be collected immediately. The primary advantages of this technique over others, such as the Bou-Rouch sampler (Bou and Rouch, 1967) are that: (i) the small size of the well minimizes disturbance of surrounding sediments and it can remain in place to allow collection of subsequent samples; (ii) the sampler does not require priming with water and as a result is fully quantitative and (iii) the sample does not pass through the mechanism of the pump and as a result specimens are less prone to damage. Each sample comprised 6 l of hyporheic water pumped from the base of the well using a bilge pump. For each hyporheic water sample, pH, conductivity, dissolved oxygen concentration, and water temperature were measured (Hanna Instruments) before passing the sample through a 90-µm mesh sieve to isolate the fauna. Benthic water characteristics (pH, conductivity, dissolved oxygen concentration and water temperature) were also recorded before the collection of faunal samples. Benthic and hyporheic invertebrate samples were preserved in the field in 4% formaldehyde, and returned to the laboratory for processing and identification. In the laboratory, invertebrate taxa were identified to species level except Baetidae (Ephemeroptera—mayfly larvae), Chironomidae (non-biting midge larvae) and Oligochaeta (worms).

To examine long-term temporal trends within the faunal data, box-plots or error bar graphs were assessed. The influence of inter-annual flow variability was investigated using the number of individuals and the number of taxa per sample (species richness). These two measures were standardized before analysis by calculating z-scores for individual sample sites from 1992 to 1999 (site mean = 0 and standard deviation = 1). This method of standardization does not alter the shape of the time-series curves at individual sites or correlation coefficients with independent variables, thus allowing comparisons between the responses of multiple sites to the same external factor (discharge variability). To examine the influence of antecedent hydrological conditions on the most common taxon recorded on the Little Stour, the amphipod shrimp *Gammarus pulex* (L.), mean annual and monthly discharge characteristics up to 12 months before sample collection were examined using scatter plots and by calculating correlation coefficients between river flow (discharge) characteristics and the standardized number of individuals per sample for the four riffle sites (also used during the 2006 study period). One way analysis of variance (ANOVA) was used to examine temporal differences between benthic and hyporheic invertebrate communities during 2006 following application of Levene's test to ensure that variances were homogeneous. Differences between individual months were examined using Tukey's *post-hoc* multiple comparisons tests to identify where significant differences occurred. All analyses were undertaken using the package SPSS (Version 15).

RESULTS

Benthic community response to inter-annual flow variability

The influence of three supra-seasonal droughts recorded during the study period (1992, 1996–1997 and 2005–2006) is clear on the long-term hydrograph of the Great Stour River (Figure 2) for which a continuous flow series is available and for which the Little Stour forms the largest tributary. The influence of the drought periods is evident for the Little Stour River between 1992 and 1999 (Figure 3a). However, the meteorological and hydrological droughts marking the onset of the supra-seasonal events (Summer, 1995 and Autumn, 2004) were characterized by relatively high discharge on several occasions due to high groundwater levels (H on Figure 2). Perennial flow was maintained throughout the Great Stour during the study period although a 1-km reach of the Little Stour was dewatered during 1991–1992 and 1996–1997 when extreme supra-seasonal groundwater drought conditions prevailed.

A total of 87 taxa from 48 families were recorded during the study period, ranging from only 42 taxa in 1992 to 60 taxa in 1995. The standardized number of individuals and species richness responded directly to changes in the discharge regime (Figure 3). Supra-seasonal drought conditions during 1992 and 1996–1997 resulted in low

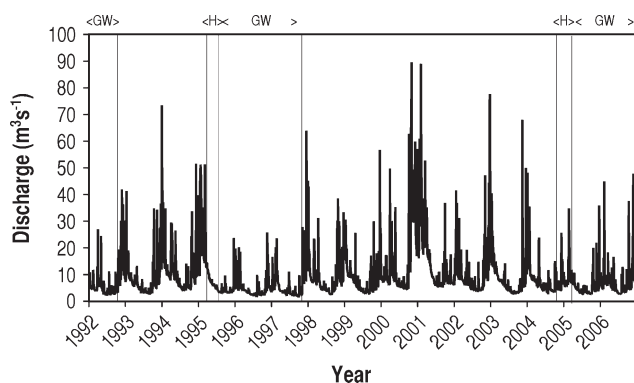


Figure 2. Hydrograph of mean daily discharge ($\text{m}^3 \text{s}^{-1}$) for the Great Stour River at Horton (1992–2006). GW indicates periods of supra-seasonal groundwater drought and H indicates the onset of meteorological and hydrological drought conditions

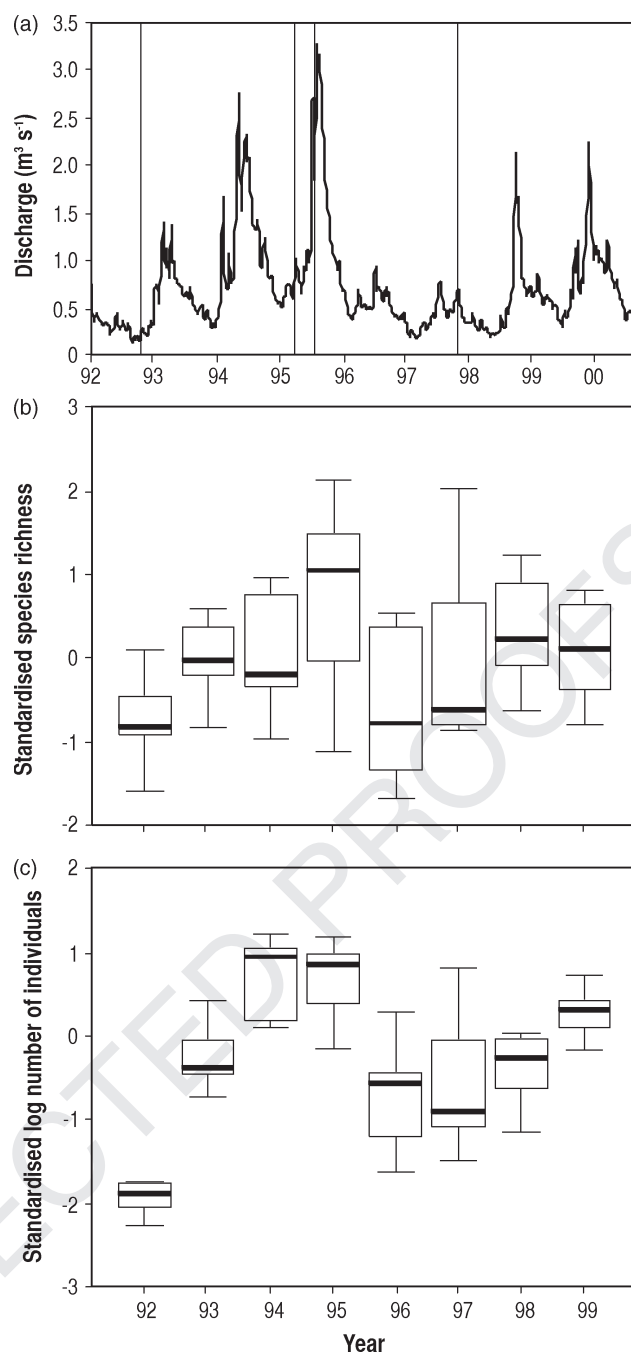


Figure 3. Time series of river flow and box-plots of macroinvertebrate assemblage indices 1992–1999 for the Little Stour River: (a) hydrograph of mean daily discharge ($\text{m}^3 \text{s}^{-1}$) for the Little Stour River at West Stourmouth—see Figure 2 for definition of vertical lines; (b) standardized species richness and (c) standardized number of individuals per sample ($\log_e$ transformed)

species richness and number of individuals per sample (Figure 3). As flow recovered following each event (1993–1994 and 1998–1999), the species richness and number of individuals per sample increased over the subsequent 2-year period (Figure 3b and c). The density of the most abundant taxon, the amphipod shrimp *G. pulex*, was significantly influenced by antecedent hydrological conditions before sampling (Table I). There was a clear positive relationship between discharge and the number of *G. pulex*, with periods of higher discharge (4–7 months before sampling) resulting in greater numbers.

Benthic and hyporheic community responses to supra-seasonal drought

The hydrological conditions recorded during 2006 resulted from an extended supra-seasonal drought that started in late 2004. As a result of low winter rainfall during 2004–2005 and 2005–2006, recharge of the chalk aquifer was limited causing an extended supra-seasonal groundwater drought (Figure 4). Above-average rainfall occurred in the catchment during May 2006 (96.4 mm) and August 2006 (111.6 mm), although the low antecedent groundwater levels precluded recovery of surface flow. The lowest river flows were recorded between August 2006 and September 2006 (Figure 4), when the riffle crests were exposed at two study sites, although flow did not cease. In addition, maximum air temperatures during July 2006 were high, resulting in elevated surface and hyporheic water temperatures (Table II). The warm mean air temperatures recorded throughout July were nationally the highest recorded in the 348-year long Central England Temperature (CET) series (Prior and Beswick, 2007).

The abundance of benthic invertebrates recorded during the study differed significantly between months ($F_{6,140} = 6.18$, $P < 0.001$) and was particularly marked by a significant reduction during July (Tukey's *post-hoc* test: all $P < 0.05$) (Figure 5a). Between April and July, the number of benthic macroinvertebrate taxa declined significantly ($F_{6,140} = 7.37$, $P < 0.001$) from an average of 23–13 taxa and was most marked during July

(Tukey's *post-hoc* test: all $P < 0.05$) (Figure 5b). This coincided with a significant reduction in the number of aquatic insect taxa ($F_{6,140} = 2.79$, $P = 0.01$) particularly mayflies (Ephemeroptera): Baetidae, *Serratella ignita* and *Caenis* spp. and caddisflies (Trichoptera): *Hydropsyche siltalai*, *Sericostoma personatum* and *Athripsodes bilineatus*. As a result, the percentage of aquatic insect larvae within the community (including mayflies, caddisflies and Diptera such as chironomid midge larvae) was significantly lower during both June and July

Table I. Pearson correlation coefficients between standardized $\log_e$ -*Gammarus pulex* and lagged discharge variables for riffles sites ($n = 4$ sites) on the Little Stour River (1992–1999)

Discharge variable	
August ($M-1$)	0.78**
July ($M-2$)	0.82**
June ($M-3$)	0.83**
May ($M-4$)	0.89**
April ($M-5$)	0.88**
March ($M-6$)	0.89**
February ($M-7$)	0.91**
January ($M-8$)	0.84**
December ($M-9$)	0.72**
November ($M-10$)	0.69**
October ($M-11$)	0.77**
September ($M-12$)	0.53*
3 months before sampling ($Y-3$)	0.47*
6 months before sampling ($Y-6$)	0.54*
9 months before sampling ($Y-9$)	0.51*
12 months before sampling ($Y-12$)	0.50*

All samples collected from last week of August to the first week of September throughout the study period. $M-n$ refers to the mean daily discharge in the month (M) before sample collection (1–12). $Y-n$ refers to the mean daily discharge in the 3, 6 and 12 months before samples collection. * $P < 0.05$; ** $P < 0.005$.

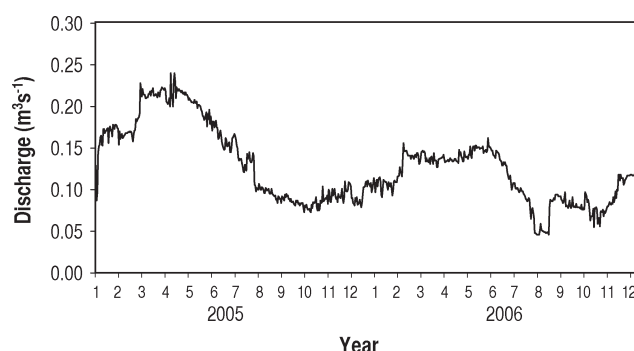


Figure 4. Hydrograph of mean daily discharge for the Little Stour River at Littlebourne (2005–2006)

(Tukey's *post-hoc* test: all $P < 0.05$) than all other months (Figure 5c).

The abundance of invertebrates within the hyporheic zone was significantly different between months ($F_{6,140} = 21.02$, $P < 0.001$). Hyporheic abundances increased significantly in July and September (Tukey's *post-hoc* test: all $P < 0.005$) but were reduced during August and October (Figure 6a). The number of taxa recorded in hyporheic samples also differed significantly between months ($F_{6,140} = 14.43$, $P < 0.001$). This was

Table II. Summary of mean monthly maximum and minimum daily temperature (with standard deviation in brackets) at Manston (Kent), and mean monthly benthic and hyporheic water temperature recorded at the study sites (April–October 2006)

	April	May	June	July	August	September	October
Max air temperature (°C)	14.2 (2.2)	13.5 (2.4)	17.2 (2.9)	21.9 (2.1)	17.7 (1.7)	18.8 (1.7)	15.1 (1.4)
Min air temperature (°C)	8.6 (2.3)	9.3 (1.9)	11.3 (2.7)	15.6 (1.7)	13.4 (1.4)	14.7 (1.7)	11.8 (2.5)
Water temperature (°C)	9.4 (0.5)	12.8 (1.8)	16.5 (2.5)	18.6 (3.1)	14.2 (1.2)	14.6 (1.1)	12.4 (0.8)
Hyporheic temperature (°C)	8.1 (0.5)	10.9 (1.7)	14.1 (2.2)	16.6 (2.1)	12.3 (1.1)	13.1 (1.0)	10.4 (0.7)

Benthic and hyporheic water samples were recorded at five locations from four sites ($n = 20$) each month.

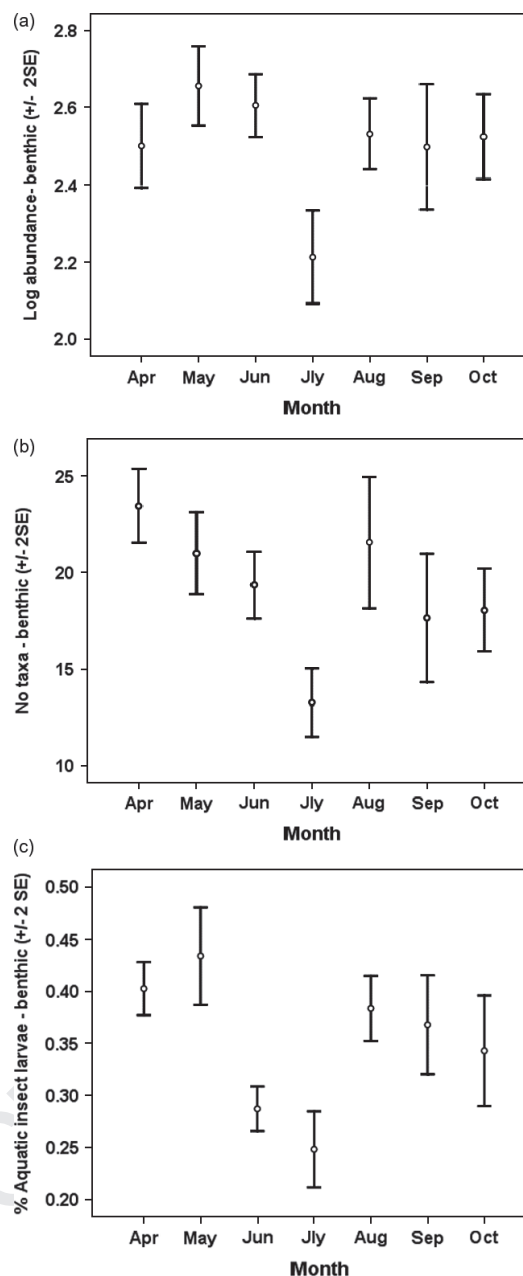


Figure 5. Little Stour benthic assemblage response (April–October 2006) during the final stages of a supra-seasonal drought event (2004–2006). Mean (± 2 standard error) of: (a) abundance of macroinvertebrates, (b) number of taxa and (c) percentage of aquatic insect larvae within samples

almost exclusively due to a significant increase in the number of taxa recorded in September (Tukey's *post-hoc* test: all $P < 0.001$) (Figure 6b) coinciding with an

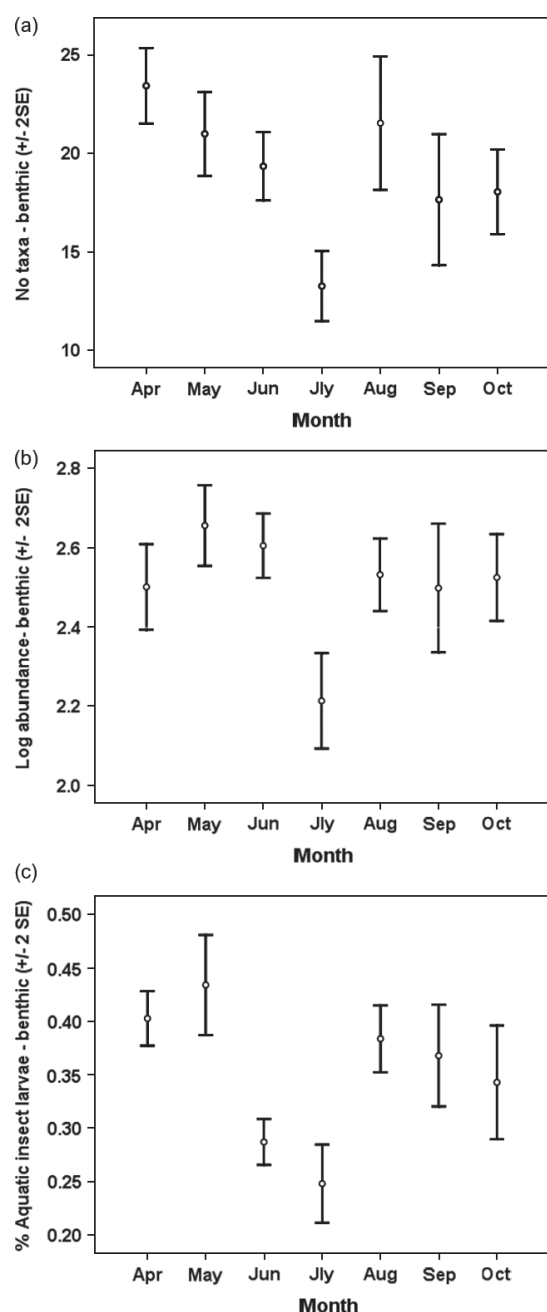


Figure 6. Little Stour hyporheic assemblage response (April–October 2006) during the final stages of a supra-seasonal drought event (2004–2006). Mean (± 2 standard error) of: (a) abundance of macroinvertebrates and (b) number of taxa

1 increase in obligate hyporheic taxa including *Proasellus*
 2 *cavaticus*, *Niphargus aquilex* and *N. fontanus*.

DISCUSSION

6 *Effects of drought on benthic and hyporheic invertebrate*
 7 *assemblages*

8 The long-term data from the Little Stour clearly
 9 demonstrates that, on an inter-annual basis, the num-
 10 ber of taxa and the number of individuals per sample
 11 appear to respond to the volume of discharge. Periods of
 12 supra-seasonal drought significantly reduced taxa rich-
 13 ness and numbers of individuals over multiple events

of varying duration. This corroborates the pattern pre-
 60 dicted for benthic fauna hypothesized by Boulton (2003),
 61 and summarized in Figure 1, although the inter-annual
 62 response of hyporheic fauna to river flow variability
 63 remains unknown. The changes in benthic taxa richness
 64 and the number of individuals per sample may reflect dif-
 65 ferences in habitat availability within the channel during
 66 drought conditions and the loss or contraction of impor-
 67 tant habitats such as clean gravels and river margin habi-
 68 tats under low-flow conditions (Harrison, 2000), and also
 69 modified life history schedules (particularly emergence of
 70 adult) of aquatic insects during extended droughts (Lake,
 71 2003).

Periods of drought-related low flows have the poten-
 73 tial to significantly modify in-stream communities in
 74 both naturally intermittent and perennial lotic systems
 75 (Boulton, 2003; Wood and Armitage, 2004; Lake, 2007).
 76 Short duration meteorological or hydrological droughts
 77 may significantly modify benthic communities in nat-
 78 urally intermittent systems (Boulton and Lake, 1992;
 79 Acuna *et al.*, 2005) but may have limited or even unde-
 80 detectable impacts within perennial streams due to the
 81 buffering effect of baseflow from groundwater sources
 82 (Wood, 1998; Humphries and Baldwin, 2003; Lake,
 83 2003). Supra-seasonal groundwater droughts also lead
 84 to significant changes in water quality (Parr and Mason,
 85 2003; Suren *et al.*, 2003), reduction of in-stream habitat
 86 availability and diversity, and changes to benthic com-
 87 munity structure and composition (Extence, 1981; Lake,
 88 2003; Dewson *et al.*, 2007), particularly if the commu-
 89 nity is not adapted to extreme low flows or drying (Lytle
 90 and Poff, 2004). However, it may take some time for
 91 the impact of drought on in-stream ecology to become
 92 apparent in groundwater-dominated systems (Wright and
 93 Symes, 1999; Wood and Armitage, 2004). Antecedent
 94 hydrological conditions are critical to determining the
 95 recession of flow during droughts (Marsh *et al.*, 2007)
 96 and this, in turn, is a primary factor influencing the ability
 97 of in-stream communities to withstand the hydrologi-
 98 cal disturbance (Humphries and Baldwin, 2003; Lake,
 99 2007).

Knowledge regarding the response of hyporheic com-
 101 munities to low flows and drought is limited and is
 102 largely confined to naturally intermittent systems where
 103 hyporheic communities have been monitored following
 104 the cessation of surface flows (Boulton and Stanley, 1995;
 105 Clinton *et al.*, 1996; del Rosario and Resh, 2000; Hose
 106 *et al.*, 2005). Only a single study has simultaneously
 107 considered the response of both benthic and hyporheic
 108 invertebrate communities to experimental low flows and
 109 this indicated no deleterious impacts on pool-dwelling
 110 invertebrates or on the abundance or vertical distribution
 111 of hyporheic macroinvertebrates as long as flow persisted
 112 (James *et al.*, 2008).

The intensive short-term study associated with the final
 114 stages of the supra-seasonal drought in 2006 suggested
 115 that the response of the benthic fauna is governed by
 116 a range of factors in addition to flow (discharge) as the
 117 lowest species richness and abundance did not correspond
 118

with the period of lowest flows. These changes coincide with the emergence of many aquatic insect taxa from the benthos and means isolating the effect of drought from natural life history characteristics is not possible unless considered alongside long-term data (Lake, 2003), which indicates that emergence of many insects occurred earlier in 2006 drought than in non-drought years. The hyporheic community responded differently to the benthos to the changes in flow and water level during the study period. Therefore, it cannot be assumed that the impact of low flow/drought upon benthic communities and the response of fauna inhabiting the hyporheic zone will be the same. This should not be unexpected because the reduction in the volume of water and the ultimate dewatering of the channel will occur within benthic habitats before water level changes within hyporheic habitats. The results of this study suggest that other abiotic parameters, such as thermal characteristics, may

be significant factors structuring both communities during supra-seasonal drought.

A conceptual model of low flow and drought influences on ecologically significant processes and interactions between the benthic and hyporheic zones

The potential influences of changes in river flow and associated abiotic factors on benthic and hyporheic communities can be hypothesized, drawing on sources from the hydrological, sedimentological and ecological literature. The conceptual model outlined in Figure 8 specifically considers the processes and interactions that may influence invertebrate communities within the hyporheic zone during periods of low flow and stream bed drying associated with surface water and groundwater drought. To our knowledge, this is the first attempt that has been made to integrate abiotic drivers with likely responses by benthic and hyporheic stream invertebrates, and provides

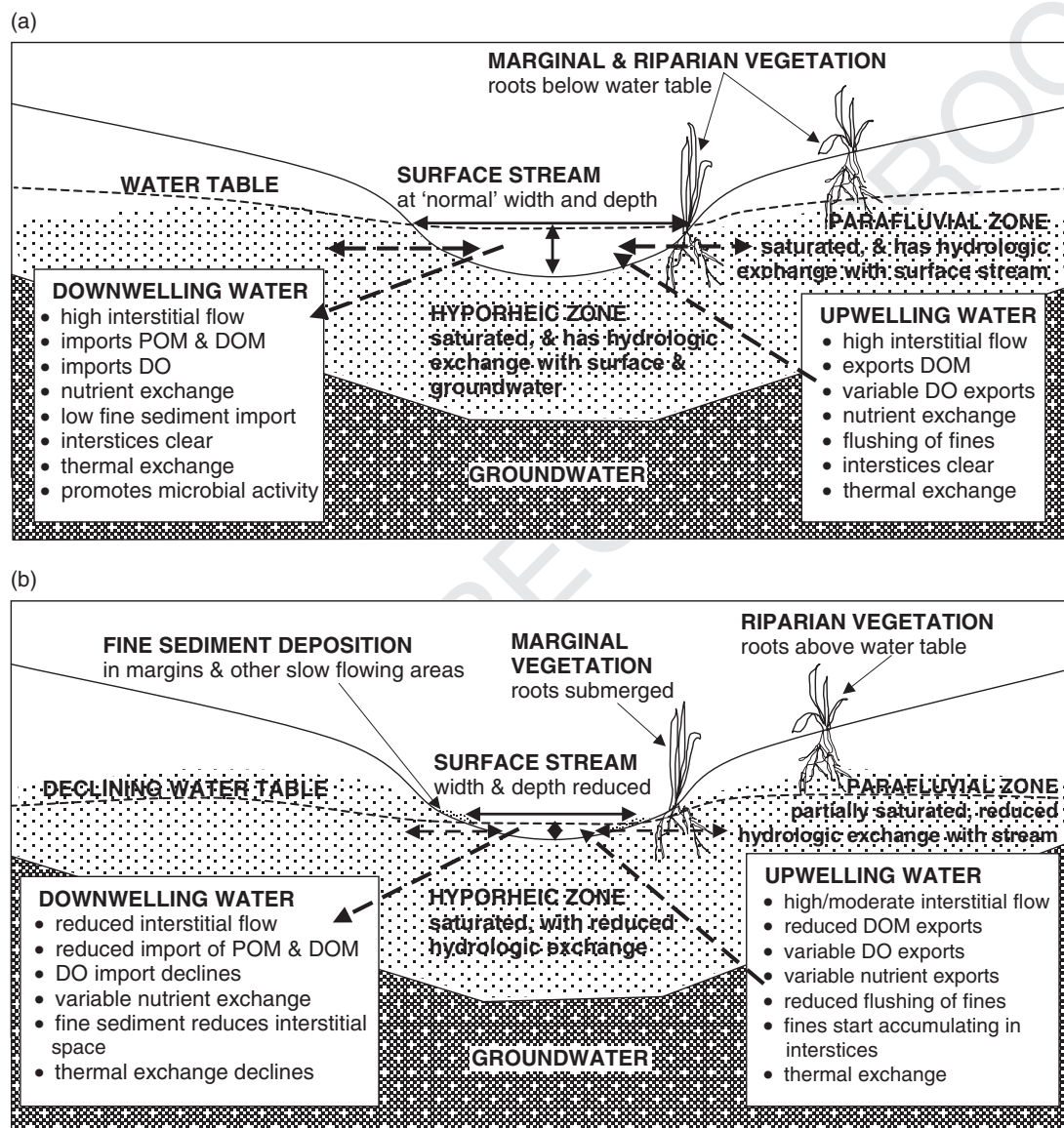


Figure 7. Conceptual model of ecologically significant processes and interactions between the benthic and hyporheic zones as a result of low flow and supra-seasonal drought: (a) unimpaired flow; (b) low/base flow; (c) loss of surface water and (d) decline of water level within the hyporheic zone

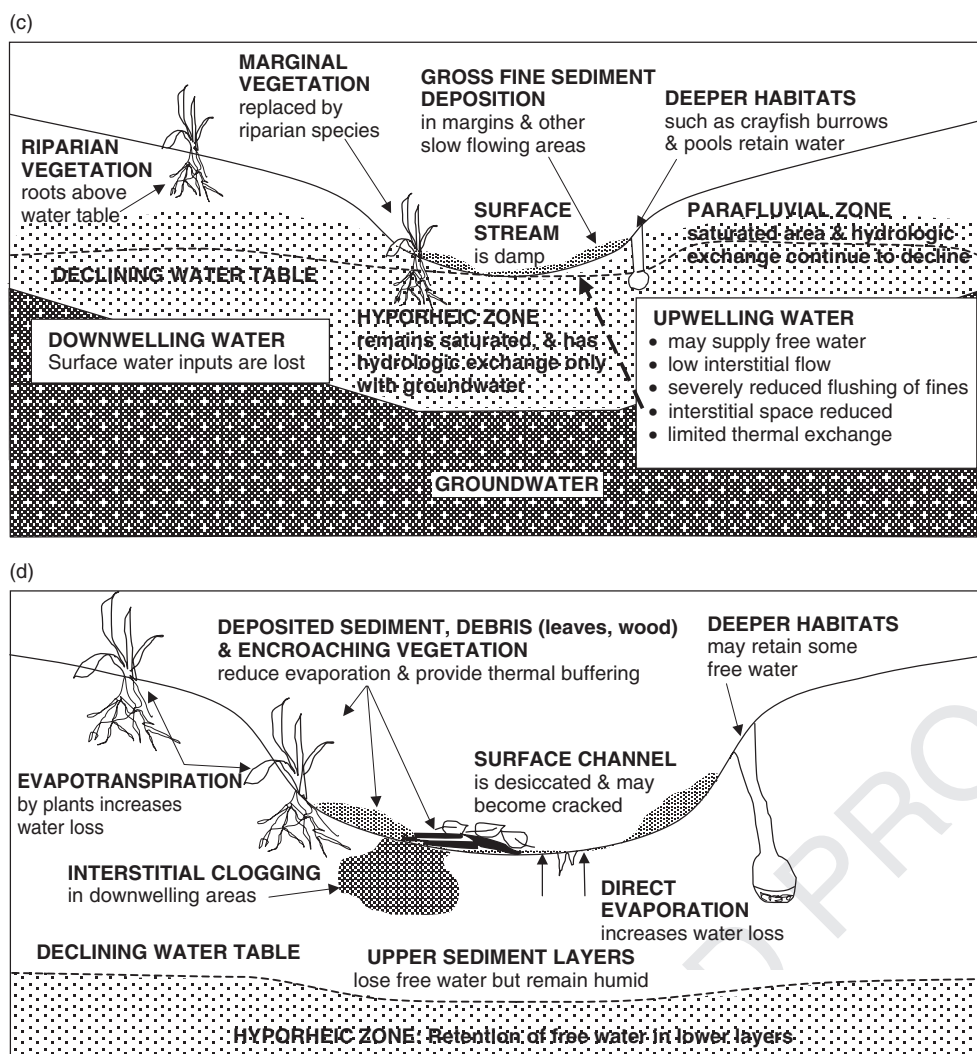


Figure 7. (Continued)

an insight into the potential impacts of anthropogenic activities on these subsystems and the hydrological linkages between them, especially during drought.

When river flow and bed integrity are unimpaired, the hyporheic zone and the adjacent parafluvial zone (*sensu* Boulton *et al.*, 1998) will be saturated, allowing both vertical and lateral hydrological exchange (Stanford and Ward, 1993; Malard *et al.*, 2002). The nature of physical and biogeochemical interactions occurring within the hyporheic zone will be strongly influenced by the direction of hydrological exchange (upwelling groundwater or downwelling surface water) and the flow velocity (Figure 7a). Local differences in the nature of these exchanges will be influenced by floodplain and channel morphology (Stanford and Ward, 1993) and at smaller scales by individual riffle, pool and bar sequences (Lefebvre *et al.*, 2006) and even individual bed elements (Boulton, 2007) which may result in micro-scale patch variability in faunal distributions (Dole-Olivier and Marmonier, 1992; Davy-Bowker *et al.*, 2006). Unimpaired hydrological exchanges within the hyporheic zone promotes thermal exchange (Hannah *et al.*, 2008), the maintenance of hyporheic interstitial permeability, porosity

and flow velocities (Malcolm *et al.*, 2005) and in-stream storage or export of nutrients (Figure 7a). As a result, the hyporheic zone may be one of the primary locations for the processing of nutrients and dissolved and particulate organic matter within some systems (Mulholland *et al.*, 2008; Pinay *et al.*, 2009) particularly through microbial activity (Hendricks, 1993; Marxsen, 2006).

As flow declines as a drought proceeds, exchange processes and connectivity between the hyporheic zone and the adjacent parafluvial will be reduced (Figure 7b). Riparian vegetation may begin to experience water stress, and marginal and in-stream vegetation will become partially or even fully exposed. Depending on whether water is locally upwelling or downwelling, the hyporheic zone may still function as a transient store or source of solutes (Stoffeth *et al.*, 2008), although the rate of exchange is likely to be significantly reduced. In the absence of flushing flows, fine sediments (<2 mm in size) may be deposited onto the bed, infiltrating and potentially clogging the interstices within the benthic and hyporheic zones (Brunke, 1999). This reduces the competency of exchange processes and the porosity and permeability of the sediments (Bo *et al.*, 2007; Meyer *et al.*, 2008),

with consequences for the supply of dissolved solutes and hyporheic oxygen (Youngson *et al.*, 2004). It also reduces living space for larger hyporheic invertebrates as well as sediment-associated benthos. The fine sediments may also be stabilized by the development of autochthonous biofilms and algal mats, further exacerbating the situation (Battin, 2000).

In many naturally intermittent rivers (in semi-arid and temperate regions) or during high magnitude supra-seasonal droughts within environments where surface flow is usually perennial, flow may almost cease and water becomes isolated within pools, although the hyporheic zone usually remains saturated (Figure 7c). As surface and groundwater levels decline, lateral interactions with the parafluvial zone may diminish or cease. Riparian and marginal vegetation typically experience significant water stress and aquatic macrophytes may be eliminated (Westwood *et al.*, 2006). Fine sediments often form a relatively impermeable crust over the substratum of the bed, beneath which anoxic conditions may exist (Smock *et al.*, 1994). Water within the hyporheic zone will continue to travel downstream and local upwelling may supply free water, maintaining a limited interstitial habitat and thermal regime within the tolerance limits of some fauna (Hose *et al.*, 2005). However, the chemical

characteristics of this hyporheic water are likely to be altered by the reduced interaction with surface waters as well as the deteriorating water quality typical of drying streams.

If drought conditions persist, levels of water within the hyporheic zone may decline, ultimately leading to the desiccation of benthic then hyporheic sediments (Figure 7d). The habitat available for aquatic organisms will become extremely limited, although refugia may exist in the form of moisture-retaining pockets of organic matter on the bed or at the margins, deeper burrows excavated by organisms such as crayfish, and hyporheic sediments that retain a high humidity (Boulton, 1989; Fenoglio *et al.*, 2006). Some aquatic taxa, particularly in systems with predictable periods of stream bed drying, display life-cycle adaptations such as diapause to withstand the desiccation (Boulton, 2003; Williams, 2006). Under extreme supra-seasonal groundwater drought conditions, exchange processes within the hyporheic zone may all cease until groundwater levels begin to recover.

A landscape perspective

Drought is a large-scale phenomenon (Lake, 2003) and when the conceptual model outlined earlier is placed in

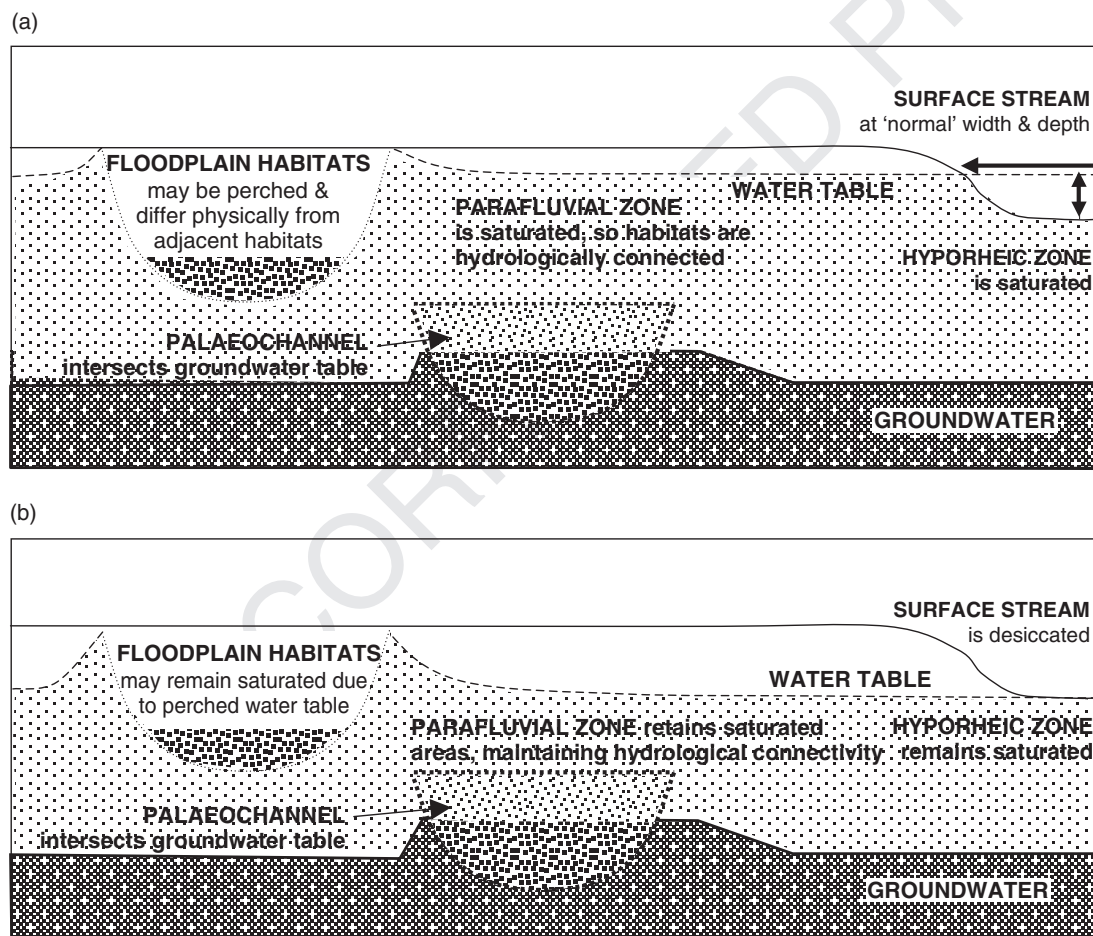


Figure 8. Conceptual model of the 'hyporheic corridor' from a landscape perspective indicating floodplain habitats such as ponds, pools, oxbow lakes and palaeochannels: (a) lateral connectivity of the hyporheic corridor during unimpaired flow and (b) lateral connectivity when surface flow in the river channel has ceased

a landscape perspective, the potential scale and significance of processes operating along the 'hyporheic corridor' (*sensu* Stanford and Ward, 1993) or within the 'stygoscape' (*sensu* Datry *et al.*, 2008) becomes apparent. The lateral connectivity of alluvial sediments and differential permeability associated with paleochannels and floodplain water bodies, such as ponds, cutoffs and backwater channels, provide corridors along which water and biota may be able to move (Figure 8). These differences in sedimentary characteristics may lead locally to elevated (perched) water tables (Malard *et al.*, 2002), which may provide small areas of surface water that persist even when flow in adjacent rivers has ceased (Figure 8b). This landscape perspective also demonstrates the refugial potential of the 'hyporheic corridor' for both hypogean and surface water fauna, respectively (Harris *et al.*, 2002). When this landscape perspective is extended to consider the wider drainage basin, the 'stygoscape' clearly extends into headwater streams and springs (Wood *et al.*, 2005) and truly subterranean habitats including cave ecosystems (Gibert and Deharveng, 2002). The potential influence of supra-seasonal groundwater drought upon subterranean ecosystems has not been widely considered to date due to the widely perceived stability of these environments and communities they support. However, the pervasive vertical hydrological linkages across the drainage basin, via hyporheic zones and shallow aquifers clearly have potential to structure communities in these habitats and affect refugial areas for surface communities. These environments and their fauna may not be so stable after all, especially during hydrological and groundwater droughts.

CONCLUSION

In-stream faunal responses to low flows and drought are frequently overlooked or only considered once the event has proceeded for many months or seasons. By which time, significant changes have often already occurred. To compound these problems, the extended and 'creeping' nature of groundwater droughts do not easily fit the timeframe of most research projects (Lake, 2003). The results of this research demonstrate the temporal impact of groundwater drought on surface and subsurface faunal assemblages at scales of individual in-stream habitats (riffles) to the landscape perspective hypothesized in our conceptual models. The research also illustrates the importance of considering lagged effects in response to hydrological inputs (precipitation) both during and following periods of drought. This is particularly important in areas subject to extended supra-seasonal groundwater droughts as the response of the aquatic faunal community is a function of the conditions within the underlying aquifer, hyporheic and parafluvial zones. In most streams, recovery of flow and the aquatic invertebrate community will only occur once the aquifer, parafluvial and hyporheic zones are fully saturated.

Until stream hydrologists, ecologists and river managers fully appreciate the interactions between groundwater, the hyporheic zone and the surface stream, our understanding of the effects of drought on microbial processes and the invertebrates inhabiting the hyporheic and benthic zones will be severely constrained. We contend that disappearance or reappearance of surface water is only part of the dynamic in-streams subject to drought and we urge further integrated research on surface and subsurface habitats to test hypotheses derived from our conceptual model. Currently, the model is a static one and as we learn more about the effects of antecedent conditions, we will be able to add the crucial temporal component that could predict the effects of 'drought history' on surface and hyporheic assemblages, with obvious implications for understanding the effects of climate change and anthropogenic modifications of flow regime.

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REFERENCES

- Acuna V, Munoz I, Giorgi A, Omella M, Sabater F, Sabater S. 2005. Drought and post drought recovery cycles in an intermittent Mediterranean stream: structural and functional aspects. *Journal of the North American Benthological Society* **24**: 919–933.
- Battin TJ. 2000. Hydrodynamics is a major determinant of streambed biofilm activity: From the sediment to the reach scale. *Limnology and Oceanography* **45**: 1308–1319.
- Bo T, Fenoglio S, Malacarne G, Pessino M, Sgariboldi F. 2007. Effects of clogging on stream macroinvertebrates: An experimental approach. *Limnologia* **37**: 186–192.
- Bonada N, Rieradevall M, Prat N, Resh VH. 2006. Benthic macroinvertebrate assemblages and microhabitat connectivity in Mediterranean-climate streams of southern California. *Journal of the North American Benthological Society* **25**: 32–43.
- Bou C, Rouch R. 1967. Un nouveau champ de recherches sur la faune aquatique souterraine. *Comptes Rendus Academie des Sciences Paris* **265**: 369–370.
- Boulton AJ. 1989. Over-summer refuges of aquatic macroinvertebrates in two intermittent streams in central Victoria. *Transactions of the Royal Society of South Australia* **113**: 22–34.
- Boulton AJ. 2003. Parallels and contrasts in the effects of drought on stream macroinvertebrate assemblages. *Freshwater Biology* **48**: 1173–1185.

- 1 Boulton AJ. 2007. Hyporheic rehabilitation in rivers: restoring vertical
2 connectivity. *Freshwater Biology* **52**: 632–650.
- 3 Boulton AJ, Fenwick G, Hancock PJ, Harvey MS. 2008. Biodiversity,
4 functional roles and ecosystem services of groundwater invertebrates.
5 *Invertebrate Systematics* **22**: 103–116.
- 6 Boulton AJ, Findlay S, Marmonier P, Stanley EH, Valett HM. 1998. The
7 functional significance of the hyporheic zone in streams and rivers.
8 *Annual Review of Ecology and Systematics* **29**: 59–81.
- 9 Boulton AJ, Stanley EH. 1995. Hyporheic processes during flooding and
10 drying in a Sonoran Desert stream. 2. Faunal dynamics. *Archiv für
11 Hydrobiologie* **134**: 27–52.
- 12 Brunke M. 1999. Colmation and depth filtration within streambeds:
13 Retention of particles in the hyporheic interstices. *International Review
14 of Hydrobiology* **84**: 99–117.
- 15 Caruso BS. 2002. Temporal and spatial patterns of extreme low flows
16 and effects on stream ecosystems in Otago, New Zealand. *Journal of
17 Hydrology* **257**: 115–133.
- 18 Clinton SM, Grimm NB, Fisher SG. 1996. Response of a hyporheic
19 invertebrate assemblage to drying disturbance in a desert stream.
20 *Journal of the North American Benthological Society* **15**: 700–712.
- 21 Datry T, Scarsbrook MS, Larned S, Fenwick G. 2008. Lateral and
22 longitudinal patterns within the stygocope of an alluvial river corridor.
23 *Fundamental and Applied Limnology* **171**: 335–347.
- 24 Davy-Bowker J, Sweeting W, Wright N, Clarke RT, Armott S. 2006.
25 The distribution of benthic and hyporheic macroinvertebrates from the
26 heads and tails of riffles. *Hydrobiologia* **563**: 109–123.
- 27 del Rosario RB, Resh VH. 2000. Invertebrates in intermittent and
28 temporary streams: Is the hyporheic zone a refuge from drying? *Journal
29 of the North American Benthological Society* **19**: 680–696.
- 30 Demuth S, Young AR. 2004. Regionalization procedures. *Hydrological
31 Drought: Process and Estimation Methods for Streamflow and
32 Groundwater, Developments in Water Science* **48**, Tallakse LM, Van
33 Lanen HAJ (eds). Elsevier: Amsterdam; 307–343.
- 34 Dewson ZL, James ABW, Death RG. 2007. A review of the conse-
35 quences of decreased flow for instream habitat and macroinvertebrates.
36 *Journal of the North American Benthological Society* **26**: 401–415.
- 37 Dole-Olivier MJ, Marmonier P. 1992. Patch distribution of interstitial
38 communities: prevailing factors. *Freshwater Biology* **27**: 177–191.
- 39 Extence CA. 1981. The effect of drought on benthic invertebrate
40 communities in a lowland river. *Hydrobiologia* **83**: 217–224.
- 41 Fenoglio S, Bo T, Bosi G. 2006. Deep interstitial habitat as a refuge for
42 *Agabus paludosus* (Fabricius) (Coleoptera: Dytiscidae) during summer
43 droughts. *Coleopterists Bulletin* **60**: 37–41.
- 44 Gibert J, Deharveng L. 2002. Subterranean ecosystems: A truncated
45 functional biodiversity. *Bioscience* **52**: 473–481.
- 46 Hannah DM, Malcolm IA, Soulsby C, Youngson AF. 2008. A compar-
47 ison of forest and moorland stream microclimate, heat exchange and
48 thermal dynamics. *Hydrological Processes* **22**: 919–940.
- 49 Harris PM, Roosa BR, Norment L. 2002. Underground dispersal by
50 amphipods (*Crangonyx pseudogracilis*) between temporary ponds.
51 *Journal of Freshwater Ecology* **17**: 589–594.
- 52 Harrison SSC. 2000. The importance of aquatic margins to invertebrates
53 in English chalk streams. *Archiv für Hydrobiologie* **149**: 213–240.
- 54 Hendricks SP. 1993. Microbial ecology of the hyporheic zone: a
55 perspective integrating hydrology and biology. *Journal of the North
56 American Benthological Society* **12**: 70–78.
- 57 Hose GC, Jones P, Lim RP. 2005. Hyporheic macroinvertebrates in riffle
58 and pool areas of temporary streams in south eastern Australia.
59 *Hydrobiologia* **532**: 81–90.
- Humphries P, Baldwin DS. 2003. Drought and aquatic ecosystems: an
introduction. *Freshwater Biology* **48**: 1141–1146.
- James BW, Dewson ZS, Death RG. 2008. Do stream invertebrates use
instream refugia in response to severe short-term flow reduction in
New Zealand streams? *Freshwater Biology* **53**: 1316–1334.
- Lake PS. 2003. Ecological effects of perturbations by drought in flowing
waters. *Freshwater Biology* **48**: 1161–1172.
- Lake PS. 2007. Flow generated disturbances and ecological responses:
floods and droughts. In *Hydroecology and Ecohydrology: Past, Present
and Future*, Wood PJ, Hannah DM, Sadler JP (eds). Wiley: Chichester;
75–92.
- Lefebvre S, Marmonier P, Peiry JL. 2006. Nitrogen dynamics in
rural streams: differences between geomorphic units. *Annales de
Limnologie—International Journal of Limnology* **42**: 43–52.
- Lytle DA, Poff NL. 2004. Adaptation to natural flow regimes. *Trends in
Ecology & Evolution* **19**: 94–100.
- Malard F, Tockner K, Dole-Olivier MJ, Ward JV. 2002. A landscape
perspective of surface-subsurface hydrological exchanges in river
corridors. *Freshwater Biology* **47**: 621–640.
- Malcolm IA, Soulsby C, Youngson AF, Hannah DM. 2005. Catchment-
scale controls on groundwater-surface water interactions in the
hyporheic zone: Implications for Salmon embryo survival. *River
Research and Applications* **21**: 977–989.
- Marsh T. 2007. The 2004–2006 drought in southern Britain. *Weather* **62**:
191–196.
- Marsh T, Cole G, Wilby R. 2007. Major droughts in England and Wales,
1800–2006. *Weather* **62**: 87–93.
- Marxsen J. 2006. Bacterial production in the carbon flow of a central
European stream, the Breitenbach. *Freshwater Biology* **51**: 1838–1861.
- Meyer EI, Niepagenkemper O, Molls F, Spanhoff B. 2008. An experi-
mental assessment of the effectiveness of gravel cleaning operations
in improving hyporheic water quality in potential salmonid spawning
areas. *River Research and Applications* **24**: 119–131.
- Monk WA, Wood PJ, Hannah DM, Wilson DA. 2008. Macroinvertebrate
community response to inter-annual and regional river flow regime
dynamics. *River Research and Applications* **24**: 988–1001.
- Mulholland PJ, Helton AM, Poole GC, Hall RO, Hamilton SK, *et al.*
2008. Stream denitification across biomes and its response to
anthropogenic nitrate loading. *Nature* **452**: 202–205.
- Parr LB, Mason CF. 2003. Long-term trends in water quality and their
impact on macroinvertebrate assemblages in eutrophic lowland rivers.
Water Research **37**: 2969–2979.
- Pinay G, O'Keefe TC, Edwards RT, Niaman RJ. 2009. Nitrate removal
in the hyporheic zone of a salmon river in Alaska. *•River Research
and Applications* DOI: 10.1002/rra.1164.
- Prior J, Beswick M. 2007. The record-breaking heat and sunshine of July
2006. *Weather* **62**: 174–182.
- Smakhtin VU. 2001. Low flow hydrology: a review. *Journal of Hydrology*
240: 147–186.
- Smakhtin VU, Schipper ELF. 2008. Droughts: The impact of semantics
and perceptions. *Water Policy* **10**: 131–143.
- Smock LA, Smith LC, Jones JB Jr, Hooper SM. 1994. Effects of drought
and hurricane on a coastal headwater stream. *Archiv für Hydrobiologie*
131: 25–38.
- Stanford JA, Ward JV. 1993. An ecosystem perspective of alluvial rivers:
connectivity and the hyporheic corridor. *Journal of the North American
Benthological Society* **12**: 48–60.
- Stofleth JM, Shields FD, Fox GA. 2008. Hyporheic and total transient
storage in small, sand-bed streams. *Hydrological Processes* **22**:
1885–1894.
- Suren AM, Biggs BJF, Duncan MJ, Bergy L, Lambert P. 2003. Benthic
community dynamics during summer low-flows in two rivers of
contrasting enrichment 2. Invertebrates. *New Zealand Journal of Marine
and Freshwater Research* **37**: 71–83.
- Suren AM, Jowett IG. 2006. Effects of floods versus low flows on
invertebrates in a New Zealand gravel-bed river. *Freshwater Biology*
51: 2207–2227.
- Tallaksen LM, van Lanen HAJ. 2004. Introduction. In *Hydrological
Drought: Process and Estimation Methods for Streamflow and
Groundwater, Developments in Water Science* **48**, Tallakse LM, Van
Lanen HAJ (eds). Elsevier: Amsterdam; 3–17.
- Tomlinson M, Boulton AJ, Hancock PJ, Cook PG. 2007. Deliberate
omission or unfortunate oversight: Should stygofaunal surveys be
included in routine groundwater monitoring programs? *Hydrogeology
Journal* **15**: 1317–1320.
- Westwood CG, Teeuw RM, Wade PM, Holmes NTH, Guyard P. 2006.
Influences of environmental conditions on macrophyte communities in
drought-affected headwater streams. *River Research and Applications*
22: 703–726.
- Williams DD. 2006. *The Biology of Temporary Waters*. Oxford University
Press: Oxford; 337.
- Wood PJ. 1998. The ecological impact of the 1995–1996 drought
on a small groundwater-fed stream. In *Hydrology in a Changing
Environment*, Vol. 1, Wheaton H, Kirby C (eds). Wiley: Chichester;
301–311.
- Wood PJ, Agnew MD, Petts GE. 2000. Flow variations and macroinver-
tebrate community responses in a small groundwater-dominated stream
in south-east England. *Hydrological Processes* **14**: 3133–3147.
- Wood PJ, Armitage PD. 2004. The response of the macroinvertebrate
community to low-flow variability and supra-seasonal drought within
a groundwater dominated stream. *Archiv für Hydrobiologie* **161**: 1–20.
- Wood PJ, Gunn J, Smith H, Abas-Kutty A. 2005. Flow permanence and
macroinvertebrate community diversity within groundwater dominated
headwater streams and springs. *Hydrobiologia* **545**: 55–64.

1	Wood PJ, Petts GE. 1994. Low flows and recovery of macroinvertebrates	Wright JF, Symes KL. 1999. A nine-year study of the macroinvertebrate	60
2	in a small regulated chalk stream. <i>Regulated Rivers-Research &</i>	fauna of a chalk stream. <i>Hydrological Processes</i> 13 : 371–385.	61
3	<i>Management</i> 9 : 303–316.	Youngson AF, Malcolm IA, Thorley JL, Bacon PJ, Soulsby C. 2004.	62
4	Wood PJ, Petts GE. 1999. The influence of drought on chalk stream	Long-residence groundwater effects on incubating salmonid redds: low	63
5	macroinvertebrates. <i>Hydrological Processes</i> 13 : 387–399.	hyporheic oxygen impairs embryo development. <i>Canadian Journal of</i>	64
6	Wright JF, Berrie AD. 1987. Ecological effects of groundwater pumping	<i>Fisheries and Aquatic Sciences</i> 61 : 2278–2287.	65
7	and a natural drought on the upper reaches of a chalk stream. <i>Regulated</i>		66
8	<i>Rivers-Research & Management</i> 1 : 145–160.		67
9			68
10			69
11			70
12			71
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